Band-1 Hadamard order 10640 – Classical Paley-I (q=10639); IN SPK via Paley-I; Gap 2052 Still OPEN

H_10640 Paley-I PASS; Paley-II FAIL/SKIP (q=5319=3*1773 NOT prime); Kronecker FAIL/SKIP (4*2660; 2660===0 mod 4 order-admissible; not_in_seeds / no H_2660); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-28

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-09-28_Band1_Hadamard_10632_PaleyI`; `2026-09-28_Band1_Hadamard_10628_PaleyI`; `2026-09-28_Band1_Hadamard_10608_PaleyI`; `2026-09-28_Band1_Hadamard_10568_PaleyI`; `2026-09-28_Band1_Hadamard_10560_PaleyI`; `2026-09-28_Band1_Hadamard_10532_PaleyI`; `2026-09-28_Band1_Hadamard_10500_PaleyI`; `2026-09-28_Band1_Hadamard_10488_PaleyI`; `2026-09-28_Band1_Hadamard_10464_PaleyI`; `2026-09-28_Band1_Hadamard_10460_PaleyI`; `2026-09-28_Band1_Hadamard_10428_PaleyI`; `2026-09-28_Band1_Hadamard_10400_PaleyI`; `2026-09-24_Band1_Hadamard_10392_PaleyI`; `2026-09-24_Band1_Hadamard_10344_PaleyI`; `2026-09-24_Band1_Hadamard_10332_PaleyI`; `2026-09-24_Band1_Hadamard_10304_PaleyI`; `2026-09-24_Band1_Hadamard_10272_PaleyI`; `2026-09-23_Band1_Hadamard_10268_PaleyI`; `2026-09-23_Band1_Hadamard_10224_PaleyI`; `2026-09-23_Band1_Hadamard_10212_PaleyI`; `2026-09-23_Band1_Hadamard_10164_PaleyI`; `2026-09-23_Band1_Hadamard_10160_PaleyI`; `2026-09-23_Band1_Hadamard_10152_PaleyI`; `2026-09-23_Band1_Hadamard_10140_PaleyI`; `2026-09-23_Band1_Hadamard_10112_PaleyI`; `2026-09-23_Band1_Hadamard_10104_PaleyI`; `2026-09-23_Band1_Hadamard_10100_PaleyI`; `2026-09-23_Band1_Hadamard_10068_PaleyI`.
Lab. `/workspace/experiments/band1/hadamard_10640_paleyI/`.
Cutoff. 2026-09-29T00:22:35Z (~8:22PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite construction of order 10640 only (verified max|HH^T-nI|=0), via classical Paley-I q=10639 (prime ===3 mod 4). 10640 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 10640: Paley-I **PASS** ($q = 10639$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL/SKIP** ($q = 5319 = 3 \cdot 1773$ NOT prime); Kronecker **FAIL/SKIP** (note $10640 = 4 \cdot 2660$; $2660 \equiv 0 \pmod 4$ order-admissible (but not_in_seeds / no H_{2660})). **10640** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK

10640+10632+10628+10608+10568+10560+10532+10500+10488+10464+10460+10428+10400+103
retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

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Abstract

Finite construction of order **10640**.

- H_{10632} = Paley-I ($q = 10639$) – classical SPK
- Paley-II for 10640: **FAIL/SKIP** ($q = 5319 = 5 \cdot 1063$ NOT prime)
- Kronecker for 10640: **FAIL/SKIP** ($4 \cdot 2660$; $2660 \equiv 2 \pmod 4$ NOT Hadamard-admissible; also not_in_seeds / no H_{2660})
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_10640.py`

Results. overall_pass=True; units 259/259; row blocks 84/84 (block size 128); wall 928.5649s; constructed max|HHˆT-nI|=0.

Cutoff: 2026-09-29T00:22:35Z (~8:22PM ET)

Arithmetic

- Target $n = 10640 = q + 1$ with $q = 10639$ prime $\equiv 3 \pmod 4$ — Paley-I eligible.
- Paley-II: $q = 5319 = 5 \cdot 1063$ NOT prime — FAIL/SKIP.
- Kronecker: $10640 = 4 \cdot 2660$; $2660 \equiv 2 \pmod 4$ order-admissible (but not_in_seeds / no H_{2660}) — FAIL/SKIP.
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_10640 Paley-I	Paley-I q=10639	0	911.0478
H_10640 Paley-II	FAIL/SKIP (q=5319=3*1773 NOT prime)	–	–

Object	Method	max|HHˆT-nI|	Wall (s)
H_10640 Kronecker	FAIL/SKIP (4*2660; 2660===0 mod 4 order-admissible; not_in_seeds / no H_2660)	–	–

SPK closure note

Seeds include prior SPK through **10632** lineage (**10632**; **10628**; **10608**; **10568**; **10560**; **10532**; **10500**; **10488**; **10464**; **10460**; **10428**; **10400**; **10392**; **10344**; **10332**; **10304**; **10272**; **10268**; **10260**; **10248**; **10244**; **10224**; **10212**; **10164**; **10160**; **10152**; **10140**; **10112**; **10104**; **10100**; **10068**; **10040**; **10008**; **9968**; **9932**; **9924**; **9908**; **9888**; **9884**; **9872**; **9860**; **9852**; **9840**; **9812**; **9804**; **9792**; **9788**; **9768**; **9744**; **9740**; **9720**; **9680**; **9644**; **9632**; **9624**; **9620**; **9588**; **9552**; **9548**; **9540**; **9512**; **9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**; **9324**; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; . . . ; **NOT** 8948/8644/8516/. . .) + verified **10464** (+ prior **10392+10344+10332+10304+10272+10268+10224+10212+10164+10160+10152+10140+10112+101**
Next constructive gap **2052** (OPEN).

Quantity	Value
10532 in SPK	**True** (Paley-I q=10639)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

q=10651->n=10652 Paley-I DIRECT q=10651 OR q=10667->n=10668 Paley-I DIRECT q=10667 OR n=2052 standard SPK constructive gap – method-attack / lit OR Track B / Object 1 meter await